

ABC COMMUNICATIONS LTD

Telco Customer Churn Analysis

Dataset Overview

Dataset Name	Telco Customer Churn Dataset
Source	Kaggle — Telco Customer Churn dataset
Number of Records	7,043 customer records
Number of Original Variables	21 variables
Unit of Observation	One row represents one customer

Data Dictionary

Column	Description	Data Type
customerID	Unique customer identifier	Text
gender	Customer gender	Categorical
SeniorCitizen	Senior citizen indicator	Numeric
Partner	Partner status	Categorical
Dependents	Dependent status	Categorical
tenure	Number of months with company	Numeric
PhoneService	Phone service status	Categorical
MultipleLines	Multiple-line phone service status	Categorical
InternetService	Internet service type	Categorical
OnlineSecurity	Online security service status	Categorical
OnlineBackup	Online backup service status	Categorical
DeviceProtection	Device protection service status	Categorical
TechSupport	Technical support service status	Categorical
StreamingTV	Streaming TV service status	Categorical
StreamingMovies	Streaming movies service status	Categorical
PaperlessBilling	Paperless billing status	Categorical

MonthlyCharges	Monthly customer charge	Numeric
PaymentMethod	Customer payment method	Categorical
Contract	Customer contract type	Categorical
Churn	Whether the customer churned	Categorical
TotalCharges	Total charges recorded for the customer	Numeric

Note: ServiceCount, Tenure Distribution, and Monthly Charges Distribution are derived/analysis fields created during the project and are not treated as original source variables

Data Quality Inspection

Check	Inspection / Treatment
Missing values	The dataset was checked using Excel filters and data inspection to identify blank or incomplete values in important analytical fields.
Duplicate customer IDs	customerID was checked for duplicates to support the assumption that each record represents one customer.
Incorrect data types	MonthlyCharges and TotalCharges required conversion/validation as numeric fields so that numerical calculations and analysis could be performed correctly.
Categorical values	Categorical fields were reviewed to preserve distinct values such as Yes, No, and No internet service. 'No internet service' was not treated as equivalent to 'No'.

Data Preparation and Transformation

Transformation	Reason
Converted charge fields to numeric	Required for numerical analysis and calculations.
Checked missing values	Ensure calculations and analysis are based on valid records.
Checked duplicate customer IDs	Support customer-level analysis and uniqueness.

Created ServiceCount	Summarize the number of selected services adopted by each customer.
Created helper calculations	Support interactive PivotTable, slicer, box plot, and correlation analysis.
Created correlation analysis	Examine relationships between selected numeric/customer characteristics.

ServiceCount Definition

ServiceCount represents the number of selected services adopted by a customer. In this analysis, the derived count is based on the following service fields: PhoneService, MultipleLines, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies. A service contributes 1 when its value is 'Yes'; otherwise it contributes 0.

Inspection Conclusion

The dataset was prepared for customer-level churn analysis by validating key fields, checking data quality, converting relevant charge fields to numeric values, preserving meaningful categorical distinctions, and creating derived/helper fields required for the dashboard and correlation analysis.